College Physics 2

ELECTRIC CHARGE

MPS Department | FEU Institute of Technology

College Physics 2

CHARGES

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OBJECTIVES

- Explain the basic concepts of charge and conservation of charges
- Differentiate the types of charging

ELECTRIC CHARGE

The word **"electric"** is derived from the Greek word ***elektron***, meaning amber.

They discovered that when they rubbed amber with wool, the amber could attract other objects.

Electricity is an invisible force in nature consisting of two electrical charges, the proton and electron



Figure 1. Amber. From.*https://phys.libretexts.org/@api/deki/files/7959/CNX_UPhysics_22_01_Amber.jpg?revision=1*



Figure 2. Wool. From. *https://images.pexels.com/photos/317333/pexels-photo-317333.jpeg?auto=compress&cs=tinysrgb&dpr=2&w=500*

*Trains, electronic gadgets, heating and cooling appliances, and many more, even our brain uses electricity to **work**.*

ELECTRIC CHARGE

Electrical charge is a property of matter that causes it to produce and experience electrical and magnetic effect
It can be represented by the symbol (Q) with a unit called ***Coulomb*** that is represented by C.

The unit of electrical charge is named after Charles Augustin De Coulomb.

Coulomb found out that each electric point charge exerts a mechanical force on the other.

The charge of an electron or proton:

$$e = \pm 1.602x10^{-19}\ C$$



Figure 3. Portrait of Charles Augustin de Coulomb.[image] Retrieve from http://en.citizendium.org/wiki/Charles-Augustin_de_Coulomb

ELECTRIC CHARGE

Plastic rods and fur are particularly good for demonstrating **electrostatics**, the interactions between electric charges that are at rest.

Benjamin Franklin suggested to calling the two kinds of charges which are ***positive*** *and* ***negative*** *charges.*

The key concept is: **Two positive charges or two negative charges repel each other. A positive and a negative charge attract each other.**

ELECTRIC CHARGE

When a material has an **excess** number of electrons, it has a *negative electrical charge* but when a material has a **deficiency** of electrons, it has *a positive electrical charge.*

PROPERTIES OF CHARGE

(1) Quantization of Charge : Electric charge can have only discrete values, that is, charge is quantized. A charge q must be an integral multiple of this basic unit. That is,

$Q = \pm ne$ where n = 1, 2, … and $e = \pm 1.602 \times 10^{-19} C$

A typical charge of a rubbed body is 10^{-8} C.

A body is having a charge of +0.32 C. How many electrons have been removed from it ?

SOLUTION

Since charge Q=+0.32 C, there electrons are removed from the body.

Let n be the no. of electrons

$$n = (0.32C)\left(\frac{1\,e}{1.602x10^{-19}C}\right) = 2x10^{18}e$$

PROPERTIES OF CHARGE

(2) Charge is always associated with mass : A charge cannot exist without mass, though a mass can exist without charge. The particles such as photon or neutrino have no (rest) mass. Hence, these particles can never have a charge.

The mass of a body (slightly) increases when it acquires a negative charge (by gaining some electrons). On the other hand, when a body acquires a positive charge (by losing some electrons), its mass (slightly) decreases.

PROPERTIES OF CHARGE

(2) Conservation of charge : *Electric charge is always conserved* in an isolated system. One object gains some amount of negative charge while the other gains an equal amount of positive charge.

(3) Invariance of charge : Numerical value of a charge is independent of the frame of reference. It means the value of charge on a body remains the same, whether it is stationary, or moving with a constant velocity or accelerating.

ELECTRIC CHARGE

Experiments in Electrostatics

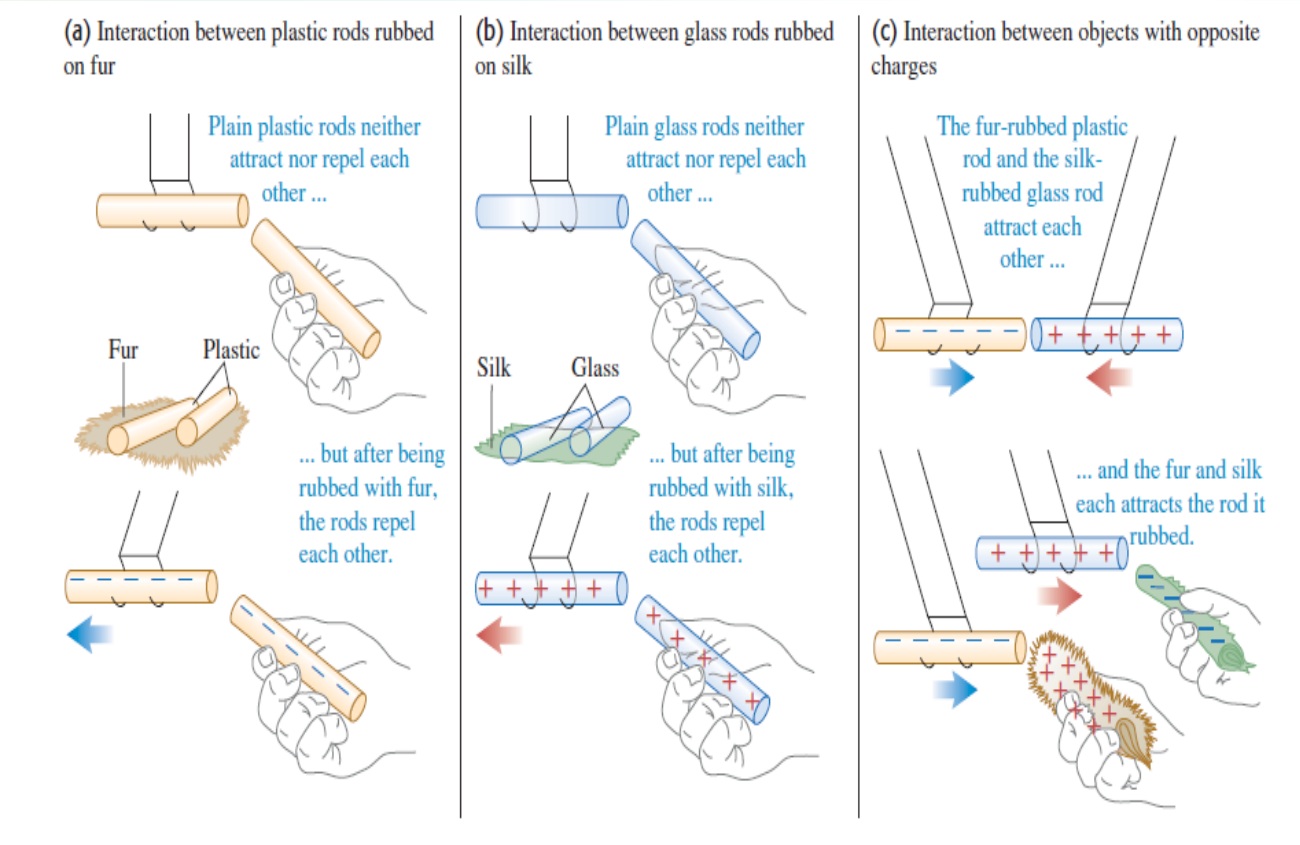


Figure 4. Experiments in Electrostatics. From. *Sear's & Zemansky's University Physics With Modern Physics* (pp 688) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

CHARGING PROCESS

CONDUCTION – charging that involves contact

- **Conductors** – materials that permit electric charge to move easily from one region of the material to another.
- **Insulators** – material that do not easily permit the electric charge to flow.
- **Semiconductor** – is a material that has an intermediate property of conductor and insulator.

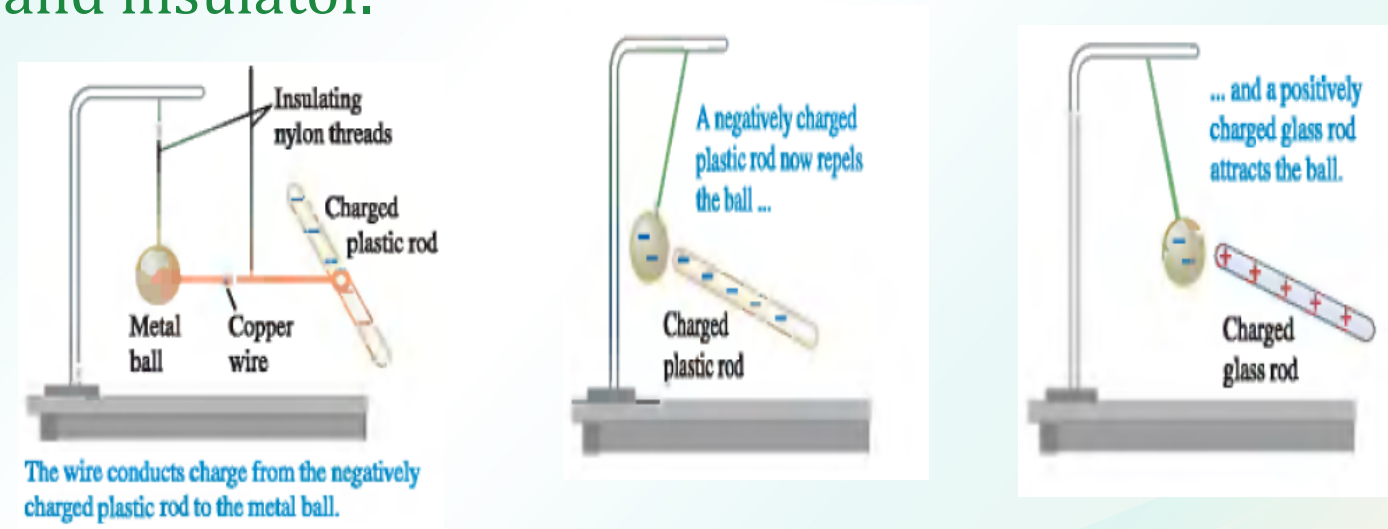


Figure 5. Charging Process From. *Sear's & Zemansky's University Physics With Modern Physics*(pp 692) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

CHARGING PROCESS

INDUCTION – Charging process that does not involve contact

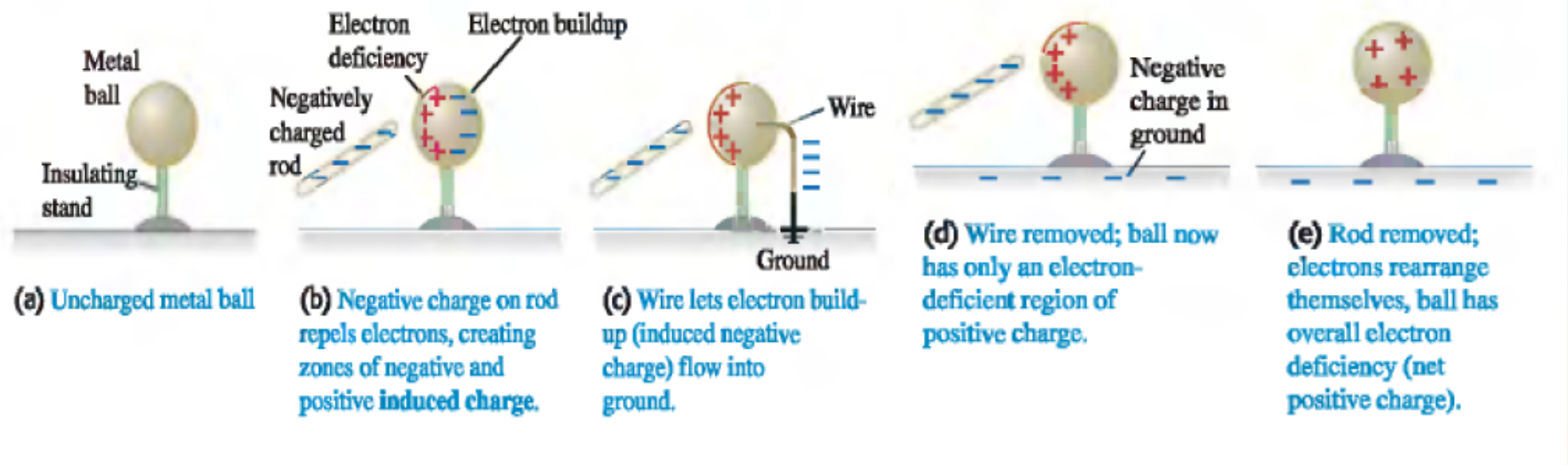


Figure 6. Charging a Metal by Induction. From. *Sear's & Zemansky's University Physics With Modern Physics*(pp 692) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

CHARGING PROCESS

FRICTION objects are rubbed with each other and charges transfer

After you electrify a comb by running it through your hair, you can pick up uncharged bits of paper with the comb. This interaction is called an ***induced*** – *charged effect.*

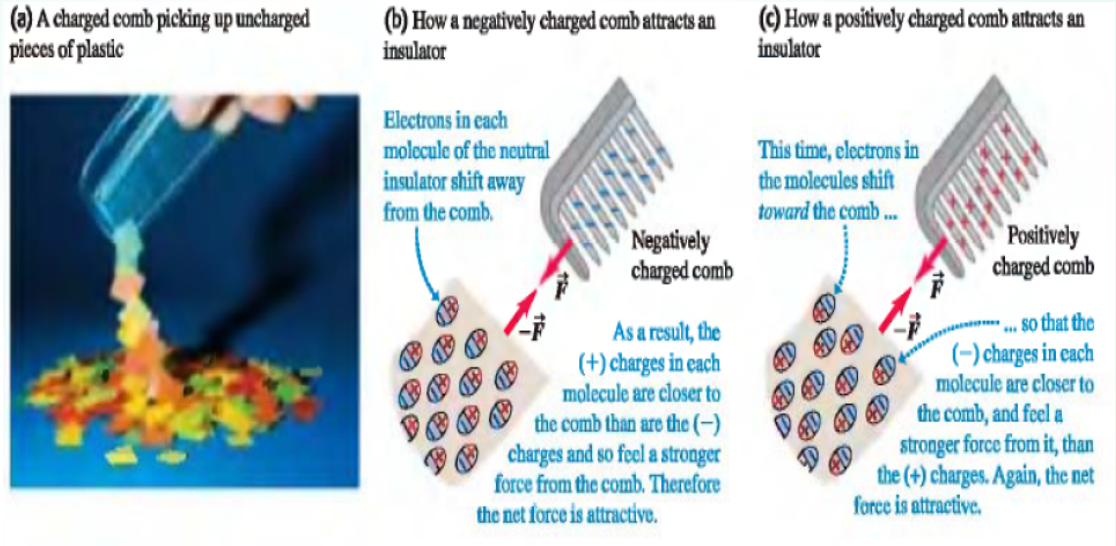


Figure 7. The charges within the molecules of an insulating material can shift slightly. From. *Sear's & Zemansky's University Physics With Modern Physics*(pp 693) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

The charges within the molecules of an insulating material can shift slightly. This is called ***Polarization*** (atomic level of separation of charges in insulators

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COLLEGE Physics 2

ELECTRIC FORCE (Coulomb's Law)

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OBJECTIVES

- Explain the basic concepts of electrostatic forces
- Be able to solve problems related to Coulomb's Law
- Solve problems related to Coulomb's Law using superposition principle

COULOMB'S LAW

"The magnitude of the electric force between two point charges is directly proportional to the product of the charges and inversely proportional to the square of the distance between them"

$$F_e = \frac{1}{4\pi\mathcal{E}_o}\frac{|q_1 q_2|}{r^2} = k_e\frac{|q_1 q_2|}{r^2}$$

Where: F – Force in Newton

$\mathcal{E}_0$ – Permittivity of free space

q_1 and q_2 – Charges in Coulomb (C)

r – distance between the charges in meters (m)

k_e – force constant (Nm^2/C^2)

$k_e = \frac{1}{4\pi\mathcal{E}_o} = 8.99 \times 10^9 \frac{Nm^2}{C^2}$

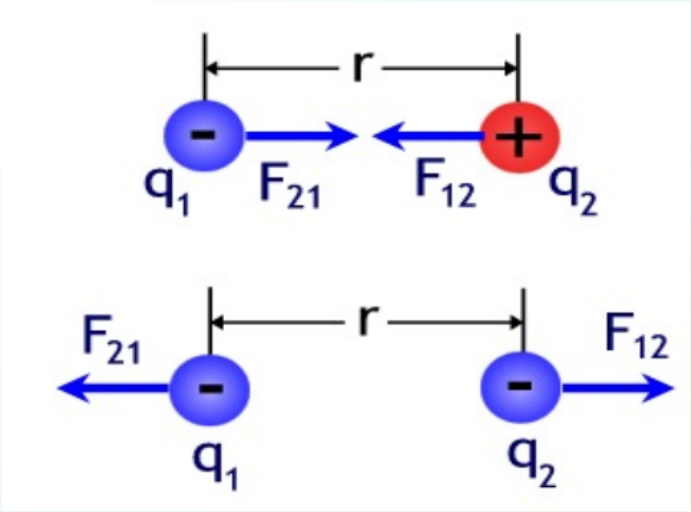


Figure 8. Tyler (2014), Coulomb's Law (image). Retrieved from http://www.wirebiters.com/electrostatics-coulombs-law/

COULOMB'S LAW

Vector form of electric force between charges q_1 and q_2:

$$\vec{F}_{12} = k_e \frac{q_1 q_2}{{r_{12}}^2} \hat{r}_{12}$$

(a) When the charges are of the same sign, the force is repulsive.
(b) When the charges are of opposite signs, the force is attractive

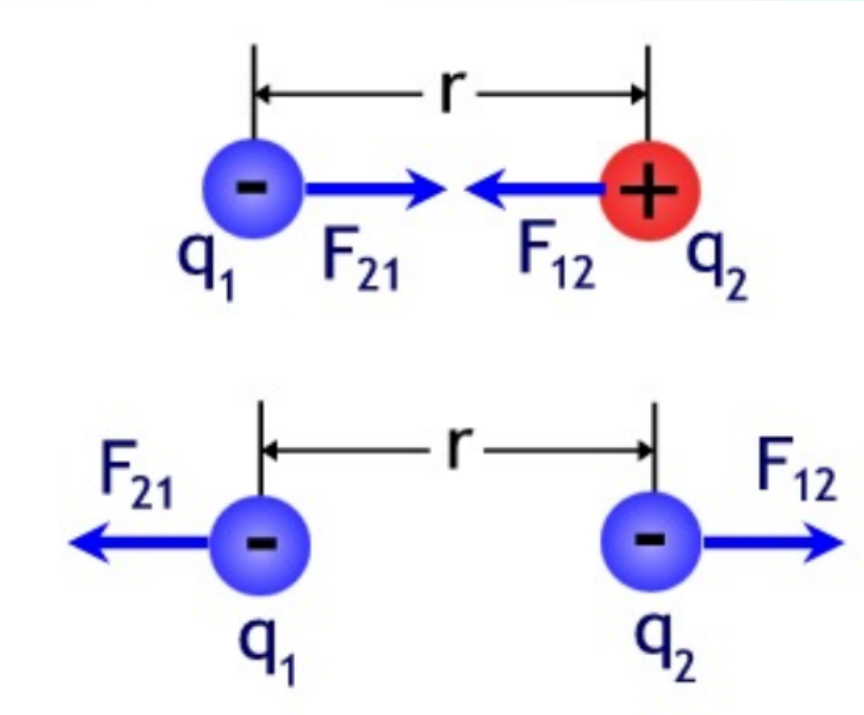


Figure 8. Tyler (2014), Coulomb's Law (image). Retrieved from http://www.wirebiters.com/electrostatics-coulombs-law/

SAMPLE

1. The Hydrogen Atom: The electron and proton of a hydrogen atom are separated by a distance of approximately 5.3×10^{-11} m. Find the magnitude of the electric force and gravitational between the two particles.

PARTICLE	CHARGE (C)	MASS (kg)
Electron (e)	-1.602×10^{-19}	9.1094×10^{-31}
Proton (p)	$+1.602 \times 10^{-19}$	1.67262×10^{-27}
Neutron	0	1.67262×10^{-27}

Table 1. Retrieved from https://profiles.uonbi.ac.ke/nyangondat/files/lesson_1_1_electrostatics_and_coulombs_law.pdf

SOLUTION

$$F_e = k_e \frac{|q_1 q_2|}{r^2} = 8.99x10^9 \, {}^{N \cdot m^2}/_{r^2} \frac{\left|(1.602x10^{-19}C)^2\right|}{(5.3x10^{-11}m)^2}$$

$$F_e = 8.2 \; x \; 10^{-8} N$$

$$F_g = G \frac{m_e m_p}{r^2} = \left(6.67x10^{-11} \, {}^{Nm^2}/_{kg^2}\right) \frac{(9.11x10^{31}kg)(1.67x10^{-27}kg)}{(5.3x10^{-11}m)^2}$$

$$= 3.6x10^{-47} N$$

Gravitational force is very small compared to the electric force between the electron and proton.

SAMPLE

2. Two point charges, q_1 = 25nC and q_2 = -75nC, are separated by a distance of 3.0 cm. Find the magnitude and direction of (a) the electric force that q_1 exerts on q_2 and (b) the electric force that q_2 exerts on q_1.

SOLUTION

(a) $F_{12} = k\frac{|q_1 q_2|}{r^2}$

$$= 8.99x10^{9\ N\cdot m^2}/_{r^2}\ \frac{|(25x10^{-9}C)(-755x10^{-9}C)|}{(3x10^{-2}m)^2}$$

(b) $F_{12} = F_{21} = 0.019N$

SUPERPOSITION PRINCIPLE

Consider more than two charges, to find the net electric force on a charge, we will be using superposition principle, that is, addition of vectors.

RECALL:

Vector form of electric force between charges q_1 and q_2:

$$\vec{F}_{12} = k_e \frac{q_1 q_2}{{r_{12}}^2} \hat{r}_{12}$$

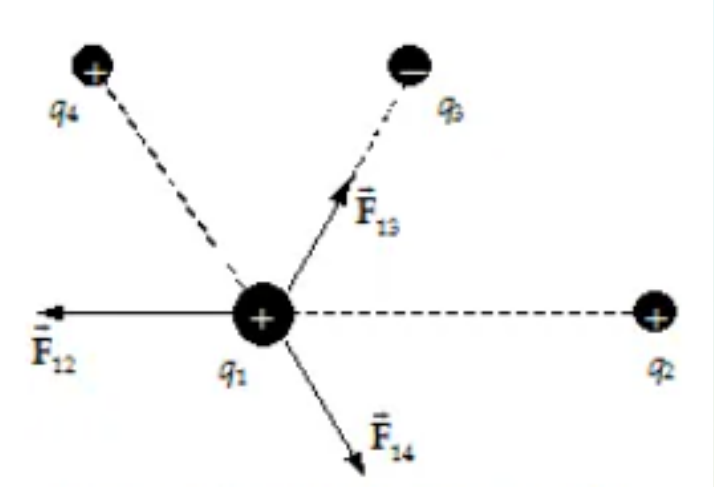


*Figure 9.*Forces acting on q_1 due to q_2, q_3, and q_4 (image). Retrieved from https://www.dronstudy.com/book/chapter-notes-electrostatics-class-12/

The net force on $\boldsymbol{q_1}$ *due to other charges:*

$$\vec{\boldsymbol{F}}_{\boldsymbol{1}} = \vec{\boldsymbol{F}}_{\boldsymbol{21}} + \vec{\boldsymbol{F}}_{\boldsymbol{31}} + \vec{\boldsymbol{F}}_{\boldsymbol{41}}$$

Where

$\vec{F}_{21}$ is the force on q_1 due to q_2

$\vec{F}_{31}$ is the force on q_1 due to q_3

$\vec{F}_{41}$ is the force on q_1 due to q_4

SAMPLE

1. Two point charges are located on the positive x-axis of a coordinate system. Charge q_1 = 1.0 nC is 2.0 cm from the origin and charge q_2 = -3.0 nC is 4.0 cm from the origin. What is the total electric force exerted on a charge q_3 = 5.0 nC located at the origin?

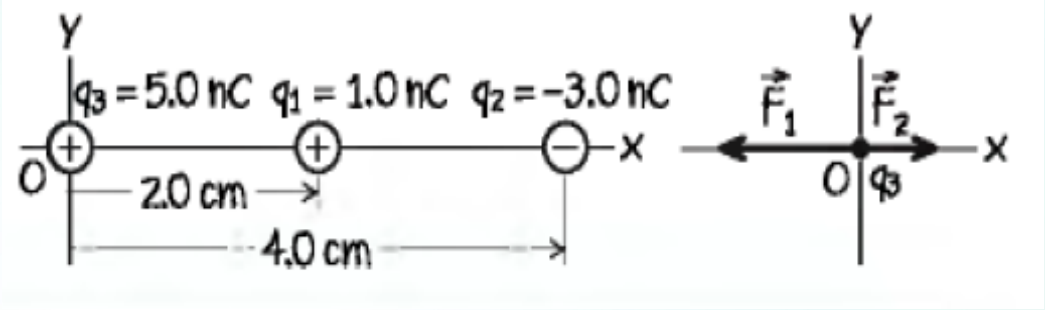


Figure 10. Two Point Charges. From. *Sear's & Zemansky's University Physics With Modern Physics*(pp 698) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson

SAMPLE

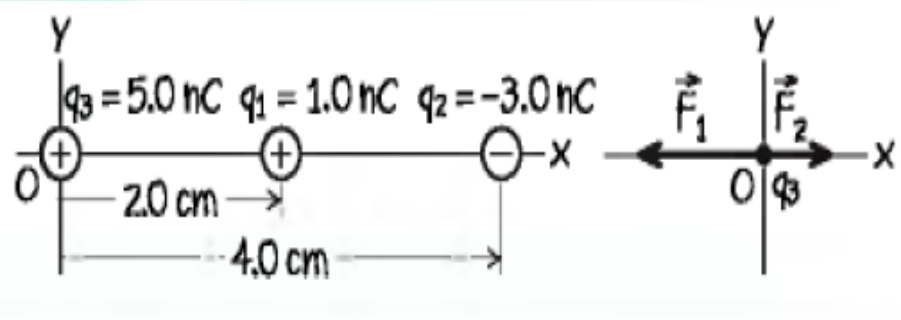


Figure 10. Two Point Charges. From. *Sear's & Zemansky's University Physics With Modern Physics*(pp 698) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson

Let $\vec{F}_1$ be the force on q_3 due to q_1 and $\vec{F}_2$ be the force on q_3 due to q_2.

$$\vec{F}_{net} = \vec{F}_1 + \vec{F}_2$$

$$\vec{F}_{net}$$

$$= -\left(8.99x10^9\ N \cdot m^2/_{r^2}\ \frac{|(5x10^{-9}C)(1x10^{-9}C)|}{(2x10^{-2}m)^2}\right)\hat{\imath}$$

$$+\left(8.99x10^9\ N \cdot m^2/_{r^2}\ \frac{|(5x10^{-9}C)(-3x10^{-9}C)|}{(4x10^{-2}m)^2}\right)\hat{\imath}$$

$$\vec{F}_{net} = -28\text{x}10^{-16}N\hat{\imath}$$

ANSWER: F_{net} = 28μN going to the left

EXERCISE

2. Four-point charges are located at the corners of a rectangle, as shown in figure. Find the net force acting on the charge q_1

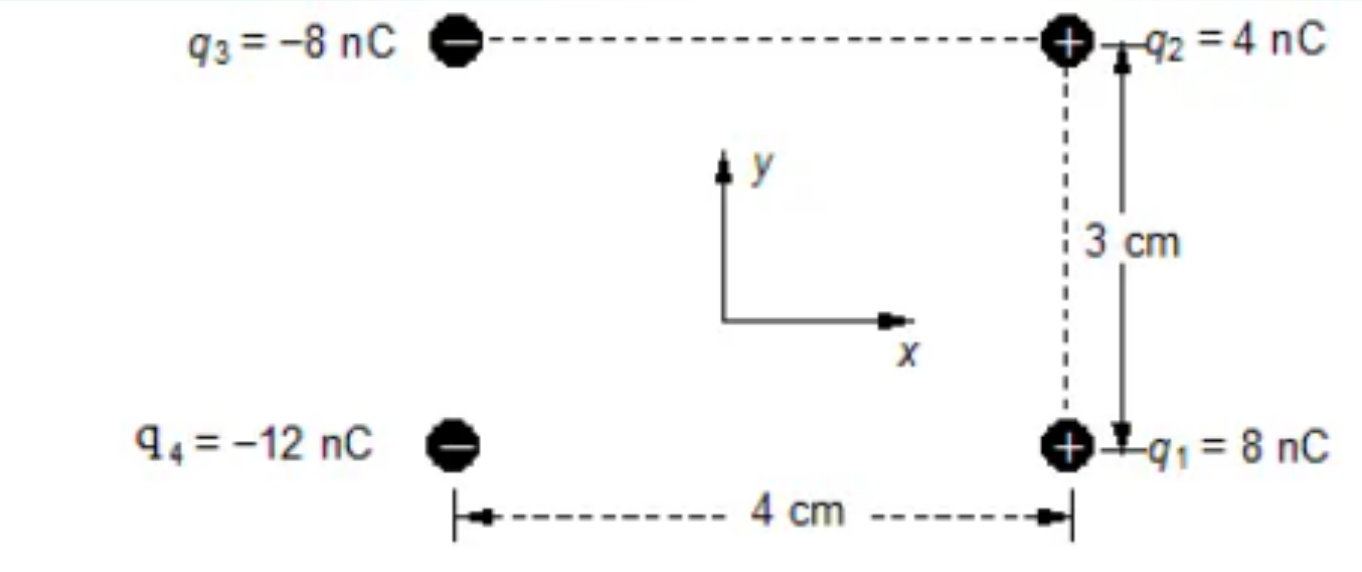


Figure 11. Four-point charges at the corners of a rectangle (image). Retrieved from https://www.dronstudy.com/book/chapter-notes-electrostatics-class-12/

EXERCISE

2. Four point charges are located at the corners of a rectangle, as shown in figure. Find the net force acting on the charge q_1

SOLUTION

The force $\vec{F}_{21}$ is repulsive, while $\vec{F}_{31}$ and $\vec{F}_{41}$ are attractive.

Find the magnitude:

$$F_{21} = k\frac{|q_1 q_2|}{{r_{12}}^2} = 9x10^9\, Nm^2/_{C^2}\,\frac{|(8x10^{-9}C)(4x10^{-9}C)|}{(3x10^{-2})^2}$$
$$= 32x10^{-5}N$$

$$F_{31} = k\frac{|q_1 q_3|}{{r_{13}}^2} = 9x10^9\, Nm^2/_{C^2}\,\frac{|(8x10^{-9}C)(8x10^{-9}C)|}{(5x10^{-2})^2}$$
$$= 23x10^{-5}N$$

$$F_{41} = k\frac{|q_1 q_2|}{{r_{12}}^2} = 9x10^9\, Nm^2/_{C^2}\,\frac{|(8x10^{-9}C)(12x10^{-9}C)|}{(4x10^{-2})^2}$$
$$= 54x10^{-5}N$$

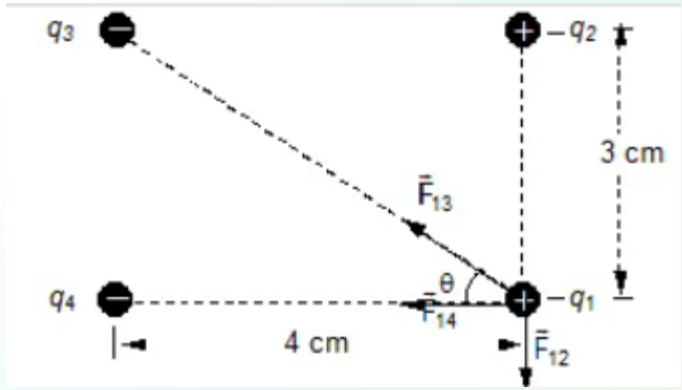


Figure 12. Four-point charges at the corners of a rectangle (image). Retrieved from https://www.dronstudy.com/book/chapter-notes-electrostatics-class-12/

EXERCISE

SOLUTION

Writing in i-j notations:

$$\vec{F} = Fcos\theta\hat{\imath} + Fsin\theta\hat{\jmath}$$

$\vec{F}_{21} = 32x10^{-5}Ncos(-90^o)\hat{\imath} + 32x10^{-5}Nsin(-90^o)\hat{\jmath}$

$\vec{F}_{21} = 32x10^{-5}N\,\hat{\jmath}$

$\vec{F}_{31} = -23x10^{-5}N\left(\frac{4}{5}\right)\hat{\imath} + 23x10^{-5}N\left(\frac{3}{5}\right)\hat{\jmath}$

$\vec{F}_{31} = -18.4x10^{-5}N\,\hat{\imath} + 13.8x10^{-5}N\,\hat{\jmath}$

$\vec{F}_{41} = 54x10^{-5}Ncos(180^o) + 54x10^{-5}Nsin(180^o)$

$\vec{F}_{41} = -54x10^{-5}N\,\hat{\imath}$

$\vec{F}_1 = \vec{F}_{21}+\vec{F}_{31}+\vec{F}_{41}$

$\vec{F}_1 = (-72.4N\hat{\imath} - 18.2Nj)x10^{-5}$

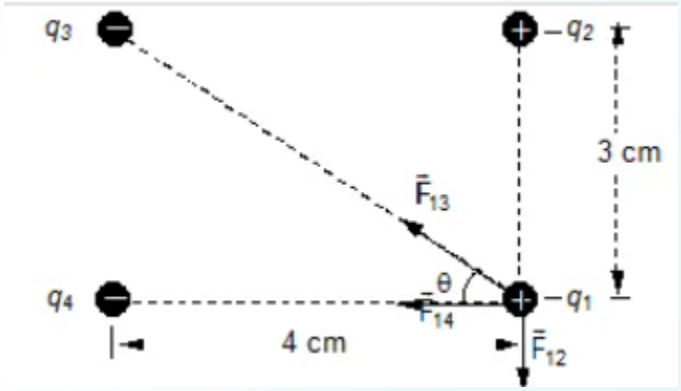


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